Fact Finding Techniques - Case Study: Library Management System

Previously, we discussed how the analyst performed the preliminary analysis. But we didn't look into the actual methods that the analyst employed to gather the information about the system. In our case the analyst used on-site observations, interviewed the staff members and used questionnaires for both staff and members of the library. Now, we will see how our analyst employed these methods.

Fact Finding Techniques

On-site Observation

Our analyst wanted to see the functioning of library. So analyst visited the library for two days and observed librarian issuing and returning books. The analyst also inspected the place where the cards are stored and from that it was seen that it was a real mess. To see if a particular book is already issued, it is a difficult and effort intensive process. The analyst also saw the records for books, members, and accounts. From site visit our analyst had a good understanding of the functioning of the system. After this, the analyst performed some personal interviews of library staff and few members. In the next section we'll look at these interviews.

Interviews

Interviews are useful to gather information from individuals. Given below is the interview between the analyst and one of the librarians, during the information gathering stage of the development of our library system.

Analyst's interview with Librarian

Analyst: Hi, I have come to talk to you regarding the functioning of your library.

Librarian: Hello, do come in. I was expecting you.

Analyst: I'll come straight to the point. Don't hesitate; you can be as much open you want. There are no restrictions.

Librarian: I'll give you my whole contribution

Analyst: Tell me are you excited about the idea of having an automated system for your library?

Librarian: Yes, I do. Very much. After all it's gonna reduce our loads of work.

Analyst: Will you elaborate on it?

Librarian: Major problem is managing the cards of members. There are so many of them. Many times cards get lost. Then we have to issue a duplicate card for it. But there is a flaw in it. It is difficult to find out if it is genuinely the case. Member can lie about it so that he/she gets an extra card. And we can't do anything about it.

Analyst: What do you think be ideal solution to this?

Librarian: There should be no cards at all. All the information should be put into computer. It'll be easy for us to check how many books we have already to a particular member.

Analyst: How often you get new members?

Librarian: Very often. At about 50 to 100 members in a month. But for two months we have freezed the membership because it is already very difficult to manage the existing 250 members. But if this whole system gets computerised then we'll open the membership. From this system, the management hopes to earn huge revenues.

Analyst: Could you explain how?
Librarian: Look every month we get about 50-100 memberships requests. After the new system is built, we will open membership to our library. There is a membership fees to be paid. Management is planning to change the membership criteria. It is planning to increase fee from 400 to 500 for half yearly and 1000 for the whole year. So in this way, we plan to get huge revenues after we have an automated system.
Analyst: Do you have different member categories?
Librarian: No, we don't have any categorisation for members. All are treated at par.
Analyst: How many books are there?
Librarian: About 5000 books
Analyst: Do you people keep records for them?
Librarian: Yes.
Analyst: Do you want facility of booking a particular title in advance?
Librarian: No we don't want any such facility. It is an overhead. So we don't have any such facility presently.
Analyst: How do you categorise your books?
Librarian: By subject.
Analyst: Would you prefer online registration for users rather than the printed form?
Librarian: Yes, we really would. Sometimes we lose these forms then we don't have any information about that particular member. It will be better to have it on computer.
Analyst: Do you have any other expectation or suggestion for the new system?
Librarian: It should be able to produce reports faster.
Analyst: Reports? I completely forgot about them. What reports you people produce presently?
Librarian: Well first is for books in the library, another for members listing, one for our current supplier of books, and reports for finance.
Analyst: Do you have some format for them?
Librarian: Yes we do have and we want that the same format be used by the new system.
Analyst: Yes we'll take care of that. Any other suggestions?
Librarian: No. You have already covered all the fields.
Analyst: Thanks for your co-operation. It was nice talking to you.
Librarian: My pleasure. Bye.
Our analyst took interviews of few members of the library in order to know about their viewpoint about the new system. One of such interview is given below.
Analyst interview with one member
Venue: Reading Room
Analyst: Hello. If you are free, I need to ask you few questions.
Member: Sure. I pleasure.
Analyst: Do you know the library people are planning to have an automated system?
Member: Yes, I do and I'm feeling good about it.
Analyst: Are you ready to pay more if there is a computerized system?

Member: In the overall functioning is going to improve then I think no one will object to paying more.
It should help us finding the books easily. But by what amount, it should matter.
Analyst: Well as far as I know they are planning to hike the membership fee from 400 to 500 for half year and 1000 for full year.
Member: That would be too much. Then in that case, they should increase the number of books to be issued. Also the number of days a book can be kept by member should also be increased.
Analyst: What you do think, how much books to be allowed for issue and for how many days.
Member: Well these people should increase number of books from 3 to at least 4. And the number of days for which a book is kept should be increased by 4 days. Presently it is for 10 days. It should be 14 days. Only then the fee hike will be justified.
Analyst: Yes, they have such plans.
Member: Then it should not bother members.
Analyst: Are you keen on online registration of members instead of normal paper one?
Member: Yes. It'll be a good practice.
Analyst: Should there be a facility to reserve a book in advance?
Member: Presently they have many copies of a single title. Usually a book is always available. I never have felt the need to reserve a book in advance.
Analyst: On what basis a book should be categorised?
Member: Well, it should be on the basis of subject.
Analyst: What do you think on what basis a search for a particular book can be done?
Member: It can be searched using subject or title.
Analyst: How often you visit this library?
Member: Daily
Analyst: Do you think magazines and cassettes should be made available in the library?
Member: I think it's a good idea.
Analyst: Do you like this library?
Member: Yes, very much. That's why I come here daily.
Analyst: Have you ever recommended this library to your friends, relatives, or to your acquaintances?
Member: Yes I very often do.
Analyst: Till now, to how many you have recommended?
Member: About 30 people.
Analyst: And how many of them have actually become its members?
Member: 25 people.
Analyst: That's really nice. People actually take your recommendation very seriously. Thank You. It was nice talking to you.
Member: Thank You.
After interviewing different people, analyst got to know about their opinions.
Questionnaires

Since the time was less it was not practical to interview every library staff. So to get the opinion of all staff, the analyst distributed questionnaires to all of them.
The questionnaire given to library staff is given below.
Instructions: Answer as specified by the format. Put NA for non-applicable situation.
1. What are your expectations out of the new system (computer based)? Rate the following on a scale of 1-4 giving a low value for low priority.
a) better cataloguing
b) better managing of users
c) better account and books management
d) computer awareness
e) any other _____
2. How many users are you expecting?
_____
3. How many books are there in library?
_____
4. How you want the books to be categorized for searching (like by title, author name or by subject)?
_____
_____
_____
5. Is there any difference in the roles (privileges) of two members?
Yes\No Please specify if Yes
_____
_____
6. Do you want facility of booking a title in advance?
Yes\No
7. Do you have data of book entered into some kind of database?
Yes\No
8. How do you want users to be categorized?
_____ or
_____
9. Would you like online registration for users rather than printed form?
Yes/No
10. Do you already have some existing categorization of books on the basis as specified in question 4
above
Yes/ No
11. Any other specific suggestion / expectation out of the proposed system.
_____
_____

Questionnaire for library staff

In order to get the views of the existing members, the analyst also distributed questionnaires to the member also. The questionnaires used by analyst for the library members are shown in figure below.

Instruction: Answer as specified by the format. Put NA for non-applicable situation.

1. Are you willing to pay extra for a library if it is fully computerized and eases finding of book, etc.

Yes\No

_____ (if Yes, how much extra are you willing to pay)

2. What you feel should be necessary for a book to be searched?

(by topic, by title, by author ,….)

_____

_____

_____

_____

3. Are you keen on online registration instead of the normal paper one.

Yes/No

4. How many titles do you feel should be issued to a single member?

_____

5. What should be the maximum duration for the issue of certain book to a member?

_____ Days.

6. Should there be a facility to reserve a book in advance?

Yes/No

7. How often do you visit the library? Choose One.

a) daily

b) Once in two days

c) Weekly

d) bi-weekly

e) Monthly

8. Should there be a facility to reserve a book on phone?

Yes/No

9. Should magazines and cassettes be included in the library

Yes/No

10. Do you recommend this library to your friends, relatives, or acquaintances?

Yes/No (if yes, to how many you recommended and out of them how many actually became the members)

Recommended :_____ Became Members : _____

Now we'll look at the techniques that the analyst employed to document the various business rules of the library.

Analyst identified the following business rules.

1) Procedure for becoming a member of library.

Anyone whose age is 18 or more than that can become a member of library. There are two types of memberships depending upon the duration of membership. First is for 6

months and other is for 1 year. 6 months membership fee is Rs 500 and 1 year membership fee is Rs 1000.
The decision tree illustrating the business rule is given below.

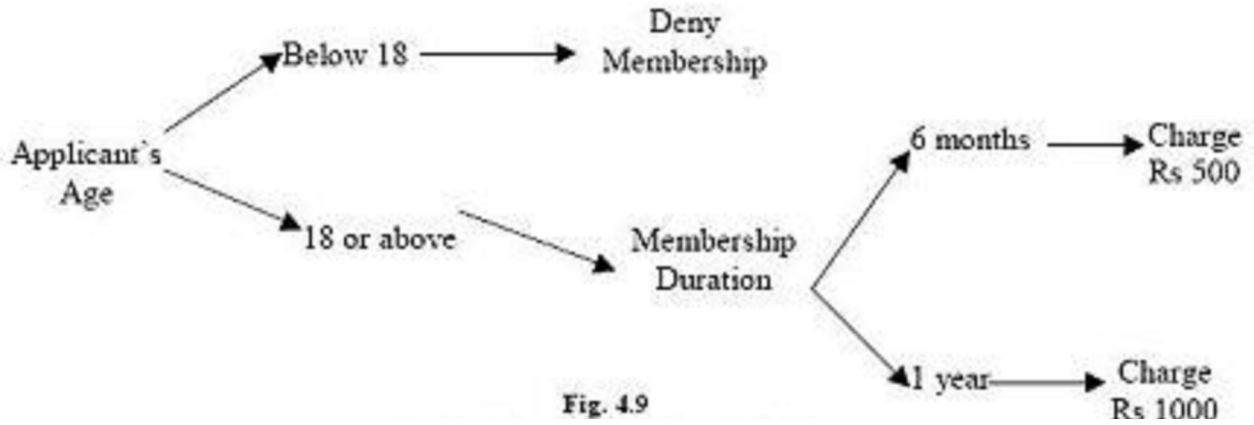
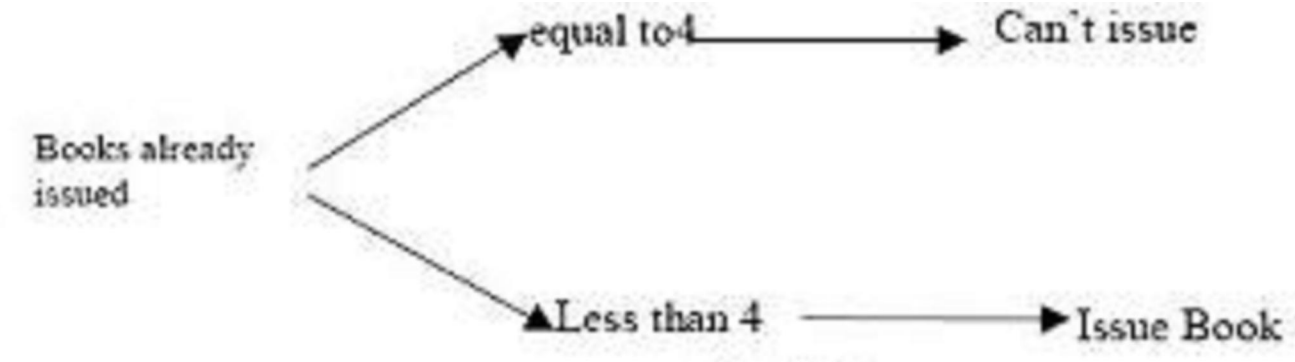


Fig. 4.9

Is Age < 18	*Y*		
Age > = 18		*Y*	*Y*
Is Memebership for 6 months ?		*Y*	
Is Memebership for 12 months ?			*Y*
Grant Membership		*X*	*X*
Deny Membership	*X*		
Charge Membership Rs. 500		*X*	
Charge Membership Rs. 1000			*X*

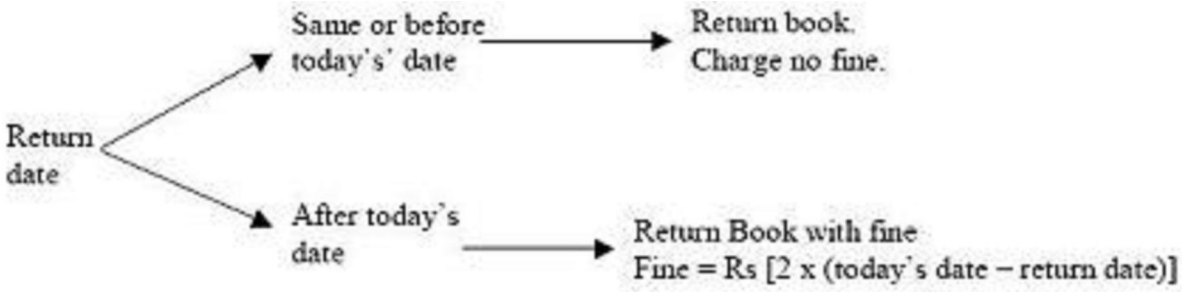
Rule for Issuing Books
If the number of books already issued is equal to 4 then no more books is issued to that member. If it is less than 4 then that book is issued.



Are Book Already Issued= 4	Y	
Are books Already issued <4		Y
Don't Issue	X	
Issue		X

Returning Books

Whenever a member returns a book, it is checked if the book is being returned after the due return date. If this is the case, then a fine of Rs 2 per day after the return date is charged. If the book is returned on the due date or before that, then no fine is charged.



Is Return Date <= Today's date	Y	
Is Return Date > Today's date		Y
Return book without charging any fine	X	
Return book with fine Fine = Rs 2 x [Today's date – Return Date]		X

Now the analyst has a good understanding of the requirements for the new system, we can move to the designing.